JAVA ENTERPRISE PRACTICALS

Practical 4: Simple Validation Using Servlet

Environment: Apache NetBeans · GlassFish Server · JDK 21

Step 1: Create the Project

1. Open Apache NetBeans.
2. Click **File** → **New Project**.
3. Select **Java Web** → **Web Application**.
4. Click **Next**.
5. Enter Project Name: **FormValidationServlet**.
6. Select your configured GlassFish Server.
7. Choose the Java EE/Jakarta EE version supported by your server.
8. Click **Finish**.

Step 2: Create index.html

1. Expand the FormValidationServlet project.
2. Right-click **Web Pages**.
3. Select **New** → **HTML File**.
4. Enter the filename **index**.
5. Click **Finish**.
6. Replace the generated code with the following:

```
<!DOCTYPE html>
<html>
<head>
    <title>Form Validation</title>
</head>
<body>
    <h2>Simple Form Validation</h2>

    <form action="validate" method="post">

        <label>Name:</label>
        <input type="text" name="name" required>
        <br><br>

        <label>Email:</label>
        <input type="text" name="email" required>
        <br><br>

        <label>Age:</label>
        <input type="number" name="age" required>
        <br><br>

        <input type="submit" value="Submit">

    </form>
</body>
</html>
```

7. Save the file using **Ctrl + S**.

Step 3: Create a Java Package

1. Right-click **Source Packages**.

2. Select **New** → **Java Package**.
3. Enter package name: `com.validation`.
4. Click **Finish**.

Step 4: Create ValidationServlet.java

1. Right-click the `com.validation` package.
2. Select **New** → **Servlet**.
3. Enter Class Name: **ValidationServlet**.
4. Enter Package: `com.validation`.
5. If the wizard asks for a URL pattern, enter `/validate`.
6. Click **Finish**.
7. Replace the generated code with the following complete program.

Note: *This code uses* `jakarta.servlet`*, suitable for GlassFish 6 or later. GlassFish 5 requires* `javax.servlet` *imports instead.*

```
package com.validation;

import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/validate")
public class ValidationServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String email = request.getParameter("email");
        String ageText = request.getParameter("age");

        boolean valid = true;

        out.println("<!DOCTYPE html>");
        out.println("<html><body>");
        out.println("<h2>Validation Result</h2>");

        // Validate name
        if (name == null || name.trim().isEmpty()) {
            out.println("<p>Name cannot be empty.</p>");
            valid = false;
        }

        // Validate email
        if (email == null ||
            !email.matches(
            "^[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+\\.[A-Za-z]{2,}$")) {

            out.println(
                "<p>Please enter a valid email address.</p>");
            valid = false;
        }
```

```
        // Validate age
        try {
            int age = Integer.parseInt(ageText);

            if (age < 18 || age > 60) {
                out.println(
                    "<p>Age must be between 18 and 60.</p>");
                valid = false;
            }

        } catch (NumberFormatException | NullPointerException e) {
            out.println(
                "<p>Age must be a valid whole number.</p>");
            valid = false;
        }

        // Display final result
        if (valid) {
            out.println(
                "<h3>Form submitted successfully!</h3>");
            out.println(
                "<p>Welcome, " + escapeHtml(name.trim()) + "!</p>");
        } else {
            out.println(
                "<h3>Validation failed. Please correct the errors.</h3>");
        }

        out.println("<p><a href='index.html'>Go Back</a></p>");
        out.println("</body></html>");
    }

    // Escape HTML characters before displaying user input
    private String escapeHtml(String value) {
        return value.replace("&", "&amp;")
                    .replace("<", "&lt;")
                    .replace(">", "&gt;")
                    .replace("\"", "&quot;")
                    .replace("'", "&#39;");
    }
}
```

8. Save the file using **Ctrl + S**.

Step 5: Verify the Project Structure

Your project should contain these files:

```
FormValidationServlet
|
|-- Web Pages
|   |-- index.html
|
|-- Source Packages
    |-- com.validation
        |-- ValidationServlet.java
```

Step 6: Build the Project

1. Right-click **FormValidationServlet**.
2. Select **Clean and Build**.
3. Wait for the build to finish.
4. If an error appears, check the NetBeans Output window.

Step 7: Run the Application

1. Right-click the project.

2. Select **Run**.
3. Wait for GlassFish to deploy the application.
4. Open `index.html` in the browser if it does not open automatically.

Typical application URL:
`http://localhost:8080/FormValidationServlet/`
Your port or context path may differ.

Step 8: Test the Application

Test 1: Valid Input

Enter:

- Name: **Rahul**
- Email: **rahul@example.com**
- Age: **22**

Expected output:

```
Validation Result

Form submitted successfully!
Welcome, Rahul!
```

Test 2: Invalid Email

Enter:

- Name: **Rahul**
- Email: **invalid-email**
- Age: **22**

Expected output:

```
Please enter a valid email address.

Validation failed. Please correct the errors.
```

Test 3: Age Below the Minimum

Enter:

- Name: **Rahul**
- Email: **rahul@example.com**
- Age: **15**

Expected output:

```
Age must be between 18 and 60.

Validation failed. Please correct the errors.
```

Test 4: Invalid Age

Enter a non-numeric age if testing a request that bypasses the browser's number-field restrictions.

Expected output:

```
Age must be a valid whole number.

Validation failed. Please correct the errors.
```

Validation Rules Summary

Field	Rule
Name	Must not be null or empty (after trimming).

Field	Rule
Email	Must match a basic email pattern (regex).
Age	Must be a whole number between 18 and 60 (inclusive).

Security note: The `escapeHtml()` helper prevents basic XSS by escaping special characters before displaying the user's name.